CATENARITY AND UNIVERSAL CATENARITY CAN BE CHECKED AT THE LOCALISATIONS AT PRIMES OR AT MAXIMAL IDEALS

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Editorial note. The counted proof refers to two same-chapter lemmas by label; both are recorded as navigation stubs pointed at their \label lines, and the arguments consuming them stay inside the counted span.

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Allowed standard prerequisites (editorial).

- Catenarity of localisations. Invoked in the counted proof at lines 25770-25771 and 25784-25785. Gives the implication from the ring to its localisations and supplies universal catenarity of a localisation of a localisation.
- Correspondence of prime chains under localisation at a maximal ideal. Used in the counted proof at lines 25774-25779. Shows chains between two primes are detected after localising at a maximal ideal containing the larger one.

Source reference. Original lemma 00NJ: sources/stacks-algebra/pinned-original-source.tex, source lines 25719–25719. Navigation pointer for \ref{lemma-localization-catenary} used inside the counted span (line 25771); the argument that consumes it is retained in the span.

Lemma 0.1. *Let R be a ring. The following are equivalent*

(1) *R is catenary,*
(2) *$R_{\mathfrak{p}}$ is catenary for all prime ideals $\mathfrak{p}$,*

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(3) $R_{\mathfrak{m}}$ *is catenary for all maximal ideals* $\mathfrak{m}$.

Assume R is Noetherian. The following are equivalent

(1) R *is universally catenary,*
(2) $R_{\mathfrak{p}}$ *is universally catenary for all prime ideals* $\mathfrak{p}$*,*
(3) $R_{\mathfrak{m}}$ *is universally catenary for all maximal ideals* $\mathfrak{m}$.

Proof. The implication (1) $\Rightarrow$ (2) follows from Lemma 00NJ in both cases. The implication (2) $\Rightarrow$ (3) is immediate in both cases. Assume $R_{\mathfrak{m}}$ is catenary for all maximal ideals $\mathfrak{m}$ of R. If $\mathfrak{p} \subset \mathfrak{q}$ are primes in R, then choose a maximal ideal $\mathfrak{q} \subset \mathfrak{m}$. Chains of primes ideals between $\mathfrak{p}$ and $\mathfrak{q}$ are in 1-to-1 correspondence with chains of prime ideals between $\mathfrak{p}R_{\mathfrak{m}}$ and $\mathfrak{q}R_{\mathfrak{m}}$ hence we see R is catenary. Assume R is Noetherian and $R_{\mathfrak{m}}$ is universally catenary for all maximal ideals $\mathfrak{m}$ of R. Let $R \to S$ be a finite type ring map. Let $\mathfrak{q}$ be a prime ideal of S lying over the prime $\mathfrak{p} \subset R$. Choose a maximal ideal $\mathfrak{p} \subset \mathfrak{m}$ in R. Then $R_{\mathfrak{p}}$ is a localization of $R_{\mathfrak{m}}$ hence universally catenary by Lemma 00NJ. Then $S_{\mathfrak{p}}$ is catenary as a finite type ring over $R_{\mathfrak{p}}$. Hence $S_{\mathfrak{q}}$ is catenary as a localization. Thus S is catenary by the first case treated above. $\square$